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Bright learn
Exercize 2
SQL Aggregate Functions & operations

SECTION 1
The Students table

Question 1

```
SELECT DISTINCT departments
FROM students;
```

departments
IT
HR
Finance

Question 2

```
SELECT department,
       AVG(age) AS avg_age
FROM Students
GROUP BY department;
```

department	avg_age
IT	20.5
HR	22
Finance	23

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Question 3

```
SELECT department,
       COUNT(*) AS student_count
FROM Students
GROUP BY department
HAVING COUNT(*)>1;
```

department	student_count
IT	2
HR	2

QUESTION 4

```
SELECT student id,
       name,
       age,
       department
FROM Students
WHERE age BETWEEN 21 AND 23
```

student_id	name	age	department
3	Charlie	21	IT
2	Bob	22	HR
5	Eve	22	HR
4	Diana	23	Finance

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Question 5

```
SELECT student_id,
       name,
       age,
       department
FROM Students
WHERE 'IT' OR 'HR" =< 21
```

student_id	name	age	departmet
2	Bob	22	HR
5	Eve	22	HR

SECTION 2

The Courses table

Question 6

```
SELECT DISSTINCT department
FROM courses
WHERE ~~total_credits~~ SUM(credits) AS total_credits <5;
```

department	total_credits
IT	11

Question 6

```
SELECT department,
       SUM(credits) AS total_credits
FROM courses
GROUP BY department
HAVING total_credits > 5
```

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department	total_credits
IT	11

Question 7

```
SELECT course_id,
       course_name,
       department,
       credits
FROM courses
WHERE credits NOT 4
```

course_id	course_name	department	credits
104	~~Finance~~ Excel	Finance	2
105	Statistics	HR	3

Question 8

```
SELECT course_id,
       course_name,
       credits
FROM courses
ORDER BY credits DESC
LIMIT 3
```

course_id	course_name	credits
102	Python	4
103	Data Science	4
101	SQL basics	3

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SECTION 3
The enrollment table

Question 9

```
SELECT MAX (grade) AS max-grade,
       MIN (grade) AS min-grade,
       AVG (grade) AS avg-grade
FROM enrollments;
```

max-grade	min-grade	avg-grade
90	78	84.6

Question 10

```
SELECT course_id,
       COUNT(*) AS enrollment_count
FROM enrollments
GROUP BY course_id;
```

course_id	enrollment_count
101	1
102	2
103	3
104	4
105	5

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SECTION 4
The Salaries table

Question 11

```
SELECT department,
       Sum (salary) AS total-salary
       SUM (bonus) AS total_bonus
FROM salaries
GROUP BY department;
```

department	total_salary	total_bonus
IT	122000	10 500
HR	109000	7500
Finance	70000	6000

Question 12

```
SELECT department,
       AVG(Salary) AS avg-salary
FROM Salaries
GROUP BY department
HAVING AVG (salary) > 55000;
```

department	avg_salary
IT	61 000
Finance	70 000

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Question 13

```
SELECT employee_id,
       name,
       Salary,
       bonus,
       (Salary + bonus) AS total_compensation
FROM Salaries
WHERE (salary + bonus) > 60000
```

employee_id	name	salary	bonus	total_compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

SECTION 5

The progects table

Question 14

```
SELECT department,
       SUM(budget) AS total_budget
       AVG(budget) AS avg_budget
FROM project
GROUP BY department
HAVING AVG(budget) > 70000
```

Department	total_budget	~~total~~ Avg_budget
IT	270000	135000
Payroll system	80000	80000

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Question 15

```
SELECT project_id
       project_name
       department,
       budget
FROM   projects
WHERE  budget BETWEEN 50000 AND 120000
AND    department NOT "Marketing';
```

Project_id	project_name	department	budget
1	AI App	IT	120000
2	Payroll system	Finance	80000
5	HR Portal	HR	50 000

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